

Methodology

Data Scientist Intern Assessment — Skynet + TOGA Intelligence

Prepared by: **Ayan Hussain**

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PART A — Explainability Questions

Sections 12 of the Assessment Brief — 10 core methodology questions

Q1 Why did you choose your risk score formula?

The formula was designed around one question: what actually prevents a cadet from completing their licence on time, safely, and without financial disruption? Each dimension in the formula maps directly to one failure mode.

Flight progress (0.25 weight) is the definitive outcome — a cadet not accumulating hours cannot graduate regardless of anything else. Study readiness (0.20) captures theoretical knowledge, which DGCA mandates as a written examination gate independent of flight hours. Quiz scores (0.15) carry independent signal from study progress — a cadet can complete all chapters passively without retaining anything; quiz performance tests actual retention. Payment risk (0.15) is a hard operational constraint — an FTO will and should halt training for cadets with unresolved balances. TOGA inactivity (0.15) is a leading indicator — cadets who disengage from ground study weeks before a milestone sortie consistently underperform. Cancellation rate (0.10) captures systemic barriers or behavioural patterns that slow progress, regardless of cause.

A transparent weighted additive formula was chosen specifically over ML because: (a) it is fully explainable to a CFI or cadet with no data science background, (b) it does not require training data, (c) every score component is directly readable from the output CSV, and (d) the assessment explicitly penalises unnecessary model complexity.

Q2 Which features had the most impact and why?

Flight progress (0.25) dominates because it is the terminal outcome. The entire purpose of the FTO is to get cadets to their required hours — every other dimension is either a leading indicator or an operational enabler of that goal.

Study readiness (0.20) ranks second because DGCA written examinations are a hard gate that no amount of flight hours can bypass. A CPL cadet with 195 hours who fails the Air Regulations exam cannot be licensed. This makes ground school performance almost as critical as flight hours.

Payment risk (0.15) and TOGA inactivity (0.15) share third place because they are both direct levers: an FTO can act on payment risk immediately (payment plans, training suspension), and TOGA inactivity can be addressed with a targeted push notification. Both drive actionable intervention within 24 hours of detection.

Cancellation rate (0.10) has the lowest weight because it is the most ambiguous — a cadet with five weather cancellations is not behaving differently from one with no cancellations. The cause matters more than the count, but this dataset does not have cadet-specific cancellation cause breakdowns.

Q3 What assumptions did you make?

- Reference date 2026-05-15 is used as "today" for all time-based calculations (inactivity, days since payment, enrollment duration).
- A cadet's historical flying rate ($\text{total_flown_hours} / \text{days_enrolled}$) is constant going forward. No acceleration or deceleration is modelled.
- The 4 cadets with payment calculation errors (C006, C023, C068, C102) are excluded from automated payment risk scoring and assigned a neutral score of 50, pending manual finance team reconciliation.
- TOGA impossible study progress rows (C005, C030: $\text{chapters_completed} > \text{total_chapters}$) are capped at total_chapters before scoring. They are flagged in the data quality report.

- Cross-base instructor deployments (20 sorties) are treated as operational decisions, not data errors — they are flagged but not excluded from analysis.
- Instructors' total_duty_hours and total_flight_hours cover the full May–June 2026 period. No daily breakdown is available.
- Cadet study readiness is computed from whichever subjects appear in toga_study.csv. If a cadet has missing subjects, the score reflects only the available data.

Q4 What data quality issue could mislead the model?

The payment calculation errors for C006, C023, C068, C102 are the most dangerous. If used as-is, incorrect outstanding amounts produce wrong payment_risk scores — potentially under-ranking genuinely at-risk cadets or over-penalising cadets whose balance was correctly entered. Our approach assigns a neutral score of 50 and flags them for manual review, but this means their true risk is unknown during the analysis window.

The second-most dangerous issue is negative aircraft utilization. Aircraft A032 had actual_end earlier than actual_start in at least one sortie record, producing a duration of -716.5 hours. If this were not caught and excluded, the aircraft utilization averages and the sortie-count-based metrics would all be corrupted silently — this is the kind of error that propagates invisibly through a pipeline.

The 4 qualification-mismatch sorties (S0044, S0107, S0184, S0263) would also mislead any model that uses sortie count as a proxy for training quality. Hours flown in a sortie where the instructor was unqualified for the aircraft type may be legally non-loggable, so including them inflates apparent flight progress.

Q5 What would you not automate or predict yet?

- **Solo sortie readiness:** The decision to send a cadet solo is a CFI judgment that involves real-time observation of aircraft handling, decision-making under pressure, and human judgment. A risk score is a decision-support input, not a go/no-go gate for solo.
- **DGCA exam pass/fail prediction:** We do not have granular chapter-level topic data, question difficulty weights, or historical exam scores. Predicting exam outcomes from quiz averages alone would produce misleading confidence.
- **Instructor performance rating:** Total flight hours and duty hours measure quantity. They tell us nothing about teaching effectiveness, cadet feedback, or skill transfer quality.
- **Weather-based cancellation prediction:** We have cancel_reason as a categorical label but no integrated MET data, METAR history, or seasonal patterns. A prediction model here would be under-specified.
- **Automated training suspension decisions:** Payment risk should trigger a human review workflow in Skynet, not an automatic block on booking. The FTO may have approved payment plans not captured in this dataset.

Q6 How would you validate this model with real FTO data?

Validation requires outcome data — which cadets actually dropped out, failed DGCA exams, or significantly delayed their training. With 12 months of historical FTO records, three validation approaches apply:

- **Historical back-test:** Apply the formula to month-3 data for each cadet cohort. Check whether cadets ranked High-risk at month 3 had 2x or greater dropout/delay rates by month 12 compared to Low-risk cadets.
- **CFI calibration (Cohen's Kappa):** Present the top 20 risk-ranked cadets to the CFI team and ask them to independently rate each cadet as Low/Medium/High based on their direct knowledge. Measure agreement between model ranking and CFI ranking. A Kappa above 0.6 indicates useful alignment.

- **Prospective tracking:** Tag current cadets with risk levels. Track outcomes over 6 months. If the model is valid, High-risk cadets should have meaningfully worse outcomes than Low-risk cadets at a statistically significant level.
- **Sensitivity analysis:** Vary each weight $\pm 10\%$ and observe rank-order stability. If one weight change dramatically reshuffles the top 20, that feature needs more careful calibration or additional data.

Q7 How would you prevent unfair ranking of cadets?

- **Separate controllable from uncontrollable cancellations:** A cadet with five Weather cancellations should not be penalised identically to one who repeatedly no-shows. The formula currently uses total cancellation rate. An improved version would use only cadet-attributable cancellations (Instructor Unavailable, no-show) and exclude external causes (Weather, ATC Restriction).
- **Course-normalise progress:** The formula uses progress_pct (flown/required), not raw hours. A PPL cadet at 80% of 45 hours and a CPL cadet at 80% of 200 hours both score identically on this dimension, which is fair.
- **Mandatory explainability:** Every score must show its component breakdown. No cadet or CFI should see only a number — they should see exactly which dimensions drove the score and by how much.
- **Human override with audit trail:** CFIs must be able to override any risk flag with documented reasoning. The override and reason must be logged so the model can be retrained with these corrections.
- **Regular weight recalibration:** Weights should be reviewed every 6 months against real outcomes. If payment risk weight consistently over-penalises cadets who catch up quickly, reduce it.
- **No peer comparisons to cadets:** Cadets should only see their own scores and progress — never how they rank against cohort peers. Competitive ranking in training contexts increases anxiety without improving learning.

Q8 How should AIRMAN display this insight without demotivating students?

The framing of risk scores to cadets is as important as the score itself. Poor framing produces anxiety and disengagement; good framing produces targeted action.

- **Frame as "support opportunity", not grade:** In TOGA, show "Your readiness is 58/100 — here's your next focus area" not "You scored 58." The score is a navigation tool, not an assessment.
- **Show trajectory, not just position:** A cadet who improved from 38 to 56 over four weeks should see that progress celebrated prominently. The trend matters more than the absolute value.
- **Never show comparative peer rankings to cadets:** Cadets see only their own data. Cohort comparisons go to the CFI dashboard only.
- **Replace vague alerts with specific actions:** "Your progress is low" is demoralising. "Complete Navigation chapters 12–20 before your Cross Country sortie scheduled on 22 May" is empowering and actionable.
- **CFI-first communication for High-risk flags:** The system notifies the instructor first. The instructor then has a human conversation with the cadet, contextualising the data and co-creating a recovery plan. No cadet should first learn of a High-risk status from a push notification.
- **Celebrate small wins:** Completing a practice test, logging a new sortie, paying an outstanding balance — all should trigger positive reinforcement signals in TOGA.

Q9 What additional data would improve your model?

- **DGCA exam scores and scheduled exam dates:** The single most valuable addition. Exam outcomes validate whether ground school performance actually predicts regulatory success.

- **METAR/TAF weather data at sortie time:** Would enable distinction between weather-caused and avoidable cancellations, and allow weather-risk modelling for sortie planning.
- **CFI assessment score per sortie (1–10 handling rating):** Currently instructors' quality of teaching is completely invisible. A per-sortie handling grade from the CFI would create the feedback loop that transforms Skynet into a training intelligence platform.
- **TOGA time-spent-per-chapter (minutes):** Chapters completed is a binary signal. Time spent reveals whether a cadet rushed through or engaged deeply. 14 chapters in 20 minutes is very different from 14 chapters in 4 hours.
- **Cadet medical certificate expiry date:** A lapsed Class 1 or Class 2 medical stops training completely. Skynet should alert 30 days before expiry — this data is currently absent.
- **Aircraft-specific cadet performance notes:** Some cadets progress faster on one aircraft type. Type-specific performance data would improve completion date projections.
- **Cadet-attributable vs external cancellation breakdown:** Currently `cancel_reason` explains the cause but not whether the cadet was responsible. Adding a `cadet_fault` boolean would fix the cancellation fairness issue in the risk score.

Q10 How would this analysis help Skynet and TOGA become more intelligent products?

For Skynet: The operations analysis shows that scheduling without embedded intelligence creates safety risks (qualification mismatches), regulatory exposure (cross-base deployments), and financial waste (fleet at under 10% average utilization). Embedding the metrics from this analysis — utilization rates, instructor workload ratios, cadet risk scores — directly into Skynet's ops dashboard transforms every scheduling decision from intuition-based to evidence-based. A dispatcher who can see "this instructor has a 0.99 flight-to-duty ratio and is operating from a non-home base" before assigning a sortie makes a fundamentally different decision.

For TOGA: The study intelligence analysis shows that TOGA already captures rich behavioural signals — quiz scores, chapter completion, practice test frequency, inactivity gaps — that are not yet being actioned. Currently TOGA is a content library. By making it a responsive platform that surfaces the right content at the right moment (based on upcoming sortie type, inactivity duration, weak subject performance), AIRMAN transforms TOGA from passive to intelligent.

The pre-sortie readiness gate (TOGA queried by Skynet at booking time) is the single highest-value integration. If a cadet books a Navigation sortie but has a study readiness score below 40 in Navigation, Skynet can flag the booking for CFI approval rather than silently confirming it. This single feature directly links ground study to airside safety — which is precisely the kind of intelligence AIRMAN's product ecosystem is designed to deliver.

PART B — Live Review Preparation

Section 19 of the Assessment Brief — 10 questions the reviewer may ask live

Q11 Explain the risk score formula.

The composite cadet risk score is a six-component weighted additive formula, scaled to 0–100:

```
risk_score = (
  0.25 x (1 - flight_progress) # flight progress shortfall
+ 0.20 x (1 - study_readiness/100) # TOGA readiness shortfall
+ 0.15 x (1 - avg_quiz_score/100) # quiz performance shortfall
+ 0.15 x payment_risk/100 # financial risk
+ 0.15 x inactivity_score # TOGA inactivity (days/30, cap 1)
+ 0.10 x cancellation_rate # sortie cancellation rate
) x 100
```

All inputs are normalized to [0,1] before weighting. The final score maps to: 0–39 = Low | 40–69 = Medium | 70–100 = High.

Each weight reflects how directly that dimension determines whether a cadet completes their licence on time. Flight progress gets the highest weight (0.25) because it is the terminal outcome. Payment gets 0.15 because it is a hard operational gate — non-payment stops training. Cancellation rate gets the lowest weight (0.10) because external causes (weather, ATC) inflate it unfairly.

Q12 Modify the risk score weights.

If a reviewer asks to change the weights, the key constraint is that all six weights must sum to 1.0. Any redistribution must be justified by domain logic. Example modifications:

- **Increase payment weight to 0.20, decrease cancellation to 0.05:** Justified if the FTO has a strict payment policy where training is suspended at a fixed outstanding threshold. Payment becomes a near-binary gate, so its predictive power increases.
- **Increase flight progress to 0.30, decrease quiz score to 0.10:** Justified for PPL cadets (45 hours only) where the written exam component is less onerous. Flight hours are the dominant constraint.
- **Increase study readiness to 0.25, decrease inactivity to 0.10:** Justified when TOGA data coverage improves to include all 7 subjects per cadet. Currently only 3 subjects are tracked, limiting study readiness signal quality.

The formula is designed to be recalibrated. After 6 months of real outcome data, weights should be adjusted using a simple linear regression of each component against actual dropout/delay outcomes — replacing domain intuition with empirical evidence.

Q13 Add one new feature to the model.

Proposed new feature: days_to_medical_expiry_risk

A cadet's medical certificate (Class 1 for CPL, Class 2 for PPL) is a legal prerequisite for all solo and cross-country flying. If a medical lapses during training, the cadet cannot fly until recertified — often a delay of 4–8 weeks. This risk is completely invisible in the current model.

```
# New feature computation:
days_to_expiry = (medical_expiry_date - today).days
medical_risk = max(0, 1 - (days_to_expiry / 90)) # 90 days = full risk
```



```
# Cap at 1.0 if already expired
```

Integration into the formula: reduce the cancellation_rate weight from 0.10 to 0.07 and add medical_risk at 0.03. This keeps the total at 1.0 and adds a real operational signal that currently has zero representation in the dataset.

Data requirement: Add a medical_expiry_date column to cadets.csv. Skynet should already hold this as part of regulatory compliance records.

Q14 Debug a wrong chart.

The most common chart bug in this analysis is the aircraft utilization chart, where A032 initially showed -389.4% utilization due to negative flight hours (-716.5 hrs). Here is the debugging process step by step:

- **Step 1 — Identify the symptom:** The chart shows a bar extending to the left (negative X). This is impossible for a utilization percentage.
- **Step 2 — Trace to the source:** The utilization formula is $\text{actual_flown_hours} / \text{total_available_hours}$. A negative numerator means actual_flown_hours is negative. This comes from summing (actual_end - actual_start) over completed sorties for A032.
- **Step 3 — Find the bad row:** Filter sorties where aircraft_id == 'A032' and status == 'completed'. Compute duration for each. One row will have actual_end before actual_start — a data entry error (e.g., AM/PM swap or date rollover not handled in parsing).
- **Step 4 — Fix in the data pipeline:** Add a validation: $\text{duration_hours} = \max(0, (\text{actual_end} - \text{actual_start}).\text{total_seconds}() / 3600)$. Flag negative durations as data_issue = 'negative_duration' and exclude from utilization calculation.
- **Step 5 — Regenerate the chart:** Re-run the chart code. All bars should now be positive. A032 will show near-0% utilization (only valid sorties counted), which is still an actionable insight.

Q15 Explain one pandas transformation line by line.

Example: Computing per-cadet cancellation rate from sorties.csv

```
sorties_grp = sorties.groupby('cadet_id')['status']
cancel_rate = sorties_grp.apply(
    lambda x: (x == 'cancelled').sum() / len(x)
).reset_index()
cancel_rate.columns = ['cadet_id', 'cancellation_rate']
```

Line 1: Group the sorties DataFrame by cadet_id and select only the status column. This creates a GroupBy object — no computation yet, just a grouping instruction.

Line 2–4: Apply a lambda function to each cadet's group of status values. Inside the lambda: (x == 'cancelled').sum() counts how many status values equal 'cancelled'. len(x) is the total number of sorties for that cadet. The ratio is the cancellation rate for that cadet.

Line 5: reset_index() converts the GroupBy result (a Series with cadet_id as the index) back into a regular DataFrame with cadet_id as a column.

Line 6: Rename the columns from the default ['cadet_id', 0] to human-readable names.

Q16 Recalculate one metric manually.

Manual recalculation: Study Readiness Score for cadet C001 (Arjun Menon)

From toga_study.csv, C001 has two subjects:

- Meteorology: chapters_completed=14, total_chapters=25, avg_quiz_score=62, practice_tests=3
- Navigation: chapters_completed=20, total_chapters=30, avg_quiz_score=78, practice_tests=4

Step 1 – avg_quiz_score across subjects:

$\text{mean_quiz} = (62 + 78) / 2 = 70.0$

Step 2 – chapter progress per subject:

Meteorology progress = $14/25 = 0.56 \rightarrow 56\%$

Navigation progress = $20/30 = 0.67 \rightarrow 67\%$

$\text{mean_progress} = (56 + 67) / 2 = 61.5\%$

Step 3 – practice test score (capped at 10 tests = full score):

$\text{mean_tests} = (3 + 4) / 2 = 3.5$

$\text{test_score} = \min(3.5 / 10, 1.0) \times 100 = 35.0$

Step 4 – weighted sum:

$\text{study_readiness} = 0.40 \times 70.0 + 0.30 \times 61.5 + 0.30 \times 35.0$

$= 28.0 + 18.45 + 10.5 = 56.95 \approx 57 / 100$

C001 scores 57/100 on study readiness — Medium range. The low practice test count (3.5 average) is the primary drag. Recommendation: attempt at least 5 practice tests per subject before the next milestone sortie.

Q17 Identify one misleading insight in your own report.

In the Skynet Operations Analysis, the headline states: **"100% of Fleet is Under-Utilised — every aircraft is below 40% utilization."**

This is technically true but potentially misleading for two reasons:

- **The dataset covers only 2.5 weeks of actual data (May 1–15) even though the sorties span May–June.** The utilization denominator (total_available_hours) appears to be a monthly figure, but the numerator (actual flown hours from sorties) only reflects 2.5 weeks of completed sorties because the reference date is May 15. If you divide 2.5 weeks of flying against a full-month availability figure, every aircraft will mechanically appear under-utilised even if operations are perfectly healthy.
- **Many aircraft simply have no sorties assigned to them in this dataset.** A fleet of 120 aircraft serving 150 cadets with 300 sorties means many aircraft never appear in sorties.csv at all — their "utilization" is 0% by absence, not by operational failure. This should have been called out explicitly.

A more accurate framing: "Of the 120 aircraft in fleet, only [N] were assigned sorties during the analysis period. Of those active aircraft, average utilization was [X]%. Fleet capacity significantly exceeds current cadet demand."

Q18 Suggest how this would become a real Skynet dashboard.

A production Skynet operations dashboard built on this analysis would have four main views:

- **Fleet Status Tile (real-time):** Aircraft status board showing each aircraft as a coloured tile — Green (flying), Amber (available but unscheduled), Red (maintenance/AOG), Grey (data error). Clicking a tile shows defect count, days since last inspection, and upcoming scheduled sorties.
- **Instructor Workload View (weekly):** Heatmap of instructor availability by base and qualification type. Overloaded instructors (flight ratio >0.85) highlighted in red. The scheduler can drag-drop sorties to rebalance load before it becomes a duty-hours compliance issue.
- **Cadet Risk Board (daily):** A traffic-light ranked list of cadets by composite risk score. Each cadet row shows: risk level, top risk reason, next scheduled sortie, and a one-click "Flag for CFI review" button. The

CFI sees this at the start of each duty day.

- **Dispatch Reliability Widget (rolling 30-day):** Completion rate, average delay, and top cancellation reason — updated daily. Allows the FTO ops manager to spot deteriorating trends (rising weather cancellations, increasing instructor unavailability) before they become systemic.

The backend would be a Python/FastAPI service reading from the same PostgreSQL database Skynet already uses, with the dashboard rendered in React. The risk score computation runs nightly as a scheduled job.

Q19 Suggest how TOGA can use this for personalisation.

TOGA's personalisation layer would use three signals from this analysis:

- **Weak subject targeting:** When a cadet opens TOGA, their home screen shows a "Focus Zone" — the single weakest subject (lowest quiz score relative to chapter progress) with a targeted 10-question drill. The cadet cannot easily navigate away without dismissing it, ensuring weak areas get addressed first rather than cadets gravitating to subjects they already know.
- **Pre-sortie brief mode:** Skynet triggers a TOGA push notification 48 hours before each scheduled sortie: "Your Navigation sortie is in 2 days — here are the 3 most-tested topics." TOGA surfaces only the 5–10 most relevant chapters for that lesson type, drawn from a mapping of lesson_type → subject → high-priority chapters.
- **Inactivity re-engagement:** At Day 5 of inactivity, TOGA sends: "You have a [lesson_type] sortie on [date]. Your [weak_subject] score is [X] — 15 minutes now could make a real difference." The message uses the cadet's actual data, making it personal rather than generic.
- **CFI view in TOGA:** A read-only instructor tab showing each cadet's readiness score, weak subjects, last active date, and practice test count. CFIs arrive at pre-flight briefings with this data already loaded — replacing anecdotal preparation checks with objective readiness indicators.

Q20 Explain what should not be automated yet.

Several decisions in aviation training require human judgment that data cannot fully replace at this stage:

- **Solo authorisation:** The go/no-go decision for a first solo is the most consequential call a CFI makes. It requires real-time observation of aircraft handling, situational awareness, and stress response under actual conditions. A risk score is an input to this conversation, never a replacement for it.
- **Training suspension for payment issues:** Automated payment-based training blocks could cut off a cadet mid-programme unfairly if a payment plan exists that isn't captured in the database. This decision must go through a human financial review.
- **Instructor-cadet pairing:** The data can flag workload imbalances, but instructor-cadet relationship quality, communication style compatibility, and specialised skill gaps are invisible to the model.
- **DGCA exam readiness certification:** An FTO must certify a cadet's readiness to sit the DGCA written exam. This involves reviewing logbooks, ground school attendance records, and CFI assessment notes — none of which are fully digitised in this dataset.
- **Safety-of-flight decisions:** Any decision where an error has safety consequences (aircraft dispatch with known defects, weather go/no-go, emergency procedures) must remain with qualified personnel. The risk model is an operations intelligence tool, not an airworthiness authority.

PART C — Risk Score Formula Reference

Quick-reference card for live review — complete formula with worked example

Component	Weight	Formula	Risk Direction
Flight Progress	0.25	1 - (flown_hrs / required_hrs)	Low progress → High risk
Study Readiness	0.20	1 - (study_readiness / 100)	Low readiness → High risk
Quiz Score	0.15	1 - (avg_quiz_score / 100)	Low score → High risk
Payment Risk	0.15	payment_risk / 100	High outstanding → High risk
TOGA Inactivity	0.15	min(days_inactive / 30, 1.0)	Inactive → High risk
Cancellation Rate	0.10	cancelled / total_sorties	High cancel → High risk
TOTAL	1.00	Multiply final sum × 100	0–39 Low 40–69 Medium 70–100 High

Worked Example — Cadet C003

Dimension	Raw Value	Normalized	Weight	Contribution
Flight Progress	12.0 / 45.0 = 26.7%	1 - 0.267 = 0.733	0.25	0.183
Study Readiness	Score = 35/100	1 - 0.35 = 0.650	0.20	0.130
Quiz Score	avg = 42/100	1 - 0.42 = 0.580	0.15	0.087
Payment Risk	Score = 68/100	0.68	0.15	0.102
TOGA Inactivity	17 days inactive	min(17/30, 1) = 0.567	0.15	0.085
Cancellation Rate	1 cancelled / 2 sorties	0.500	0.10	0.050
TOTAL			1.00	0.637 × 100 = 63.7
RISK LEVEL				MEDIUM (40–69)

C003 scores 63.7/100 — Medium risk. Primary drivers: very low flight progress (26.7% of PPL), TOGA inactivity (17 days), and moderate quiz score (42). The cadet has only completed 2 sorties with 1 cancellation. Recommended action: CFI check-in, targeted Navigation revision on TOGA, payment plan review for outstanding ₹90,000 balance.